Finite and Clopen Primitive Sets Are Sets of Synthesis for Locally Elliptic Weighted Orlicz Algebras

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Abstract

Question 8.12 of Carter's paper asks for geometric conditions on subsets of the primitive and coadjoint-orbit spectra which imply spectral synthesis for Hermitian weighted Orlicz algebras on unipotent p-adic groups. We prove two such conditions. Every clopen primitive subset is synthetic for every locally elliptic group covered by Carter's hypotheses. Every finite closed set of admissible primitive points is also synthetic; for a p-adic unipotent group, CCR makes every primitive point admissible. Consequently, every finite or clopen subset of the p-adic coadjoint-orbit space is a set of synthesis. The clopen proof uses central idempotents in compact-open corners and approximates them by exact finite-Hecke Riesz projections in Carter's minimal ideal $j(C)$. This cannot extend to all compact closed sets: Malliavin's theorem already supplies compact nonsynthesis sets for the unweighted algebra on the abelian group $U_2(k) \cong (k, +)$.

Status. Candidate substantial partial result, likely valid, subject to human review. It supplies the broad geometric criteria "finite" and "clopen" for the full nonabelian class in Question 8.12. Infinite closed sets with nonempty boundary remain unclassified, and compactness alone is provably insufficient.

1 Source question

The source is Max Carter, *Weighted Orlicz ∗-algebras on locally elliptic groups*, arXiv:2503.20735v2, Question 8.12 on page 29 [1]. For a p-adic unipotent matrix group N, Carter identifies

$$\mathrm{Prim}_*(L^\Phi(N,\omega)) \cong \mathfrak{n}^*/\,\mathrm{Ad}^*(N)$$

and asks which geometric properties of C imply $\ker(C) = j(C)$.

2 Definitions and setup

Let G be locally elliptic, let (Φ, Ψ) be a complementary pair of Young functions with $\Phi \in \Delta_2$, and let ω be an L^Φ-weight with $1/\omega \in L^\Psi(G)$. Put

$$\mathcal{A} = L^\Phi(G, \omega)$$

and assume that $\mathcal{A}$ is Hermitian. Carter identifies irreducible ∗-representations of $\mathcal{A}$ with irreducible unitary representations of G. For closed $C \subseteq \mathrm{Prim}_*(\mathcal{A})$ he defines

$$\ker(C) = \bigcap_{P \in C} P$$

Hermitian (which is the case if ω is sub-additive, for example) and suppose that I is a closed ideal of $L^{\Phi}(N,\omega)$ with $\mathrm{hull}_*(I) := C \subseteq \mathrm{Prim}_*(L^{\Phi}(N,\omega))$. Let $j(C)$ be the ideal of $L^{\Phi}(N,\omega)$ as defined in Theorem 7.1. Then, the ideal I must sit between $\ker(C)$ and $j(C)$ i.e. $j(C) \subseteq I \subseteq \ker(C)$. In particular, understanding the closed ideals of $L^{\Phi}(G,\omega)$ with hull equal to C would require one to understand the ideal theory of the algebra $\ker(C)/j(C)$.

The set $C \subseteq \mathrm{Prim}_*(L^{\Phi}(N,\omega))$ is called a ***set of synthesis*** if $\ker(C)/j(C)$ is trivial, or equivalently, $\ker(C)$ is the only closed ideal of $L^{\Phi}(N,\omega)$ with hull C. The property of $L^{\Phi}(N,\omega)$ being Wiener is equivalent to the empty set being a set of synthesis, so under our assumption that $L^{\Phi}(N,\omega)$ is Hermitian, it follows that the empty set is a set of synthesis by Theorem 1.3. Also, by the previous corollary, we may identify C with a subset of $\mathfrak{n}^*/\mathrm{Ad}^*(N)$. We thus pose the following problem.

Question 8.12. *Let N be a group of n-dimensional unipotent matrices over a p-adic field. Let (Φ,Ψ) be a complementary pair of Young functions and ω an L^{Φ}-weight on N with $1/\omega \in L^{\Psi}(N)$ and $L^{\Phi}(N,\omega)$ Hermitian. Given*

$$C \subseteq \mathfrak{n}^*/\mathrm{Ad}^*(N) \cong \mathrm{Prim}_*(L^{\Phi}(N,\omega)),$$

can geometric properties of C, when viewed as a subset of $\mathfrak{n}^*/\mathrm{Ad}^*(N)$, be linked to algebraic properties of the algebra $\ker(C)/j(C)$? In particular, what geometric properties of C imply that it is a set of synthesis?

Figure 1: Question 8.12, rendered from page 29 of arXiv:2503.20735v2.

and, using a smooth functional calculus, constructs a closed two-sided ideal $j(C)$ with $j(C) \subseteq \ker(C)$ and $\mathrm{hull}_*(j(C)) = C$ [1, Theorem 7.1]. The set C is a *set of synthesis* if $j(C) = \ker(C)$.

Write $\mathcal{D}(G)$ for the compactly supported locally constant functions. For a compact-open subgroup $H \le G$, put

$$e_H = \mu(H)^{-1} 1_H.$$

Then $e_H = e_H^* = e_H * e_H$, and $\pi(e_H)$ is projection onto the H-fixed vectors of a unitary representation π.

3 Proof intuition

Every algebraic test function belongs to a finite-dimensional Hecke algebra. If it vanishes under the representations in C, then so does its central support projection. Carter's smooth functional calculus recovers that projection exactly, placing the test function in $j(C)$.

For a clopen C, fix H and pass to the unital corner $e_H \mathcal{A} e_H$. The clopen split of its primitive spectrum produces a central self-adjoint idempotent p_H which is zero on C and one on the complementary primitive points. The idempotent need not itself be a test function. Approximate it in the Orlicz norm by self-adjoint Hecke functions and take their Riesz projections for the spectral cluster near one. Those projections remain in finite Hecke algebras, vanish exactly on C, and converge to p_H; hence $p_H \in j(C)$. Every H-smoothed element of $\ker(C)$ is multiplied identically by p_H, so compact-open smoothing completes the proof.

For a finite family of admissible representations, the same smoothing makes the evaluation range finite dimensional. A bounded linear lift corrects dense Hecke approximants to vanish exactly at every chosen point.

4 The algebraic test-function lemma

Lemma 1 (Dense locally finite test algebra)**.** *$\mathcal{D}(G)$ is dense in $\mathcal{A}$, and every $d \in \mathcal{D}(G)$ belongs to a finite-dimensional convolution $*$-subalgebra of $\mathcal{D}(G)$.*

Proof. The Δ_2 hypothesis gives density of compactly supported functions in the weighted Orlicz space. A locally elliptic group has a compact-open basis, so uniform approximation on compact sets by locally constant step functions, together with local boundedness of ω, gives density of $\mathcal{D}(G)$.

The compact support of d lies in a compact-open subgroup K. Uniform local constancy on K gives an open subgroup under whose left and right translations d is invariant. Replacing it by its normal core gives an open normal $L \triangleleft K$. The L-bi-invariant functions supported in K form a finite-dimensional convolution $*$-algebra, isomorphic after Haar normalization to $\mathbb{C}[K/L]$, and contain d. □

Lemma 2 (Algebraic kernel elements lie in $j(C)$)**.** *For every closed $C \subseteq \mathrm{Prim}_*(\mathcal{A})$,*

$$\mathcal{D}(G) \cap \ker(C) \subseteq j(C).$$

Proof. Put $d \in \mathcal{D}(G) \cap \ker(C)$ in a finite-dimensional convolution $*$-algebra B using Lemma 1. In its faithful regular representation, B is a finite-dimensional C^*-algebra. Let p_d be the central support projection of d, so $p_d * d = d$.

For every π whose kernel lies in C, the ideal $\ker(\pi|_B)$ contains d, hence contains p_d. Choose $0 < a < 2\pi$ with $a\|p_d\|_1 \leq 1$. Carter's Proposition 5.5 gives $\varphi \in A_\gamma$ which is zero near 0 and satisfies $\varphi(a) = 1$. Then $\varphi\{ap_d\} \in m(C)$, while the defining power series and $p_d * p_d = p_d$ give

$$\varphi\{ap_d\} = \varphi(a)p_d = p_d.$$

Thus $p_d \in j(C)$ and $d = p_d * d \in j(C)$. □

5 Clopen synthesis

Lemma 3 (The clopen corner idempotent)**.** *Let $C \subseteq \mathrm{Prim}_*(\mathcal{A})$ be clopen and $H \leq G$ compact-open. In the corner*

$$\mathcal{A}_H = e_H * \mathcal{A} * e_H$$

there is a central self-adjoint idempotent p_H such that, for every irreducible π with $\pi(e_H) \neq 0$,

$$\pi(p_H) = \begin{cases} 0, & \ker \pi \in C, \\ \pi(e_H), & \ker \pi \notin C. \end{cases}$$

Proof. The corner is unital with identity e_H. It is Hermitian: if $x = x^* \in \mathcal{A}_H$, then $x + (1 - e_H)$ is self-adjoint in the unitization of $\mathcal{A}$; compressing its nonreal resolvents shows that x has real spectrum in the corner. It is semisimple because $\mathcal{A} \subseteq L^1(G)$ embeds faithfully in the group C^*-algebra, and so does the corner.

The standard idempotent-corner correspondence, obtained by compressing irreducible representations to $\pi(e_H)\mathcal{H}_\pi$, identifies $\mathrm{Prim}_*(\mathcal{A}_H)$ with

$$U_H = \{\ker \pi \in \mathrm{Prim}_*(\mathcal{A}) : \pi(e_H) \neq 0\}.$$

Hence $Y = C \cap U_H$ and $U_H \setminus C$ are complementary closed subsets of $\mathrm{Prim}_*(\mathcal{A}_H)$. Put

$$I = \ker_{\mathcal{A}_H}(Y), \qquad J = \ker_{\mathcal{A}_H}(U_H \setminus C).$$

Semisimplicity gives $I \cap J = 0$. If $I + J$ were proper, a maximal left ideal containing it would produce a simple $\mathcal{A}_H$-module. Hermitianness makes that module unitarizable, so its primitive kernel would belong to both Y and its complement, a contradiction. Thus $I + J = \mathcal{A}_H$.

Write $e_H = p_H + q_H$ with $p_H \in I$ and $q_H \in J$. Since products between I and J lie in their zero intersection, p_H and q_H are complementary central idempotents. The ideals are $*$-closed, and uniqueness of the decomposition gives $p_H = p_H^*$. Its represented values are as stated. □

Lemma 4 (The corner idempotent belongs to $j(C)$)**.** *For C and p_H as above, $p_H \in j(C)$.*

Proof. Let $\mathcal{D}_H = e_H * \mathcal{D}(G) * e_H$. It is dense in $\mathcal{A}_H$. Choose self-adjoint $d_n \in \mathcal{D}_H$ with $d_n \to p_H$ in the $\mathcal{A}$-norm. Every d_n belongs to a finite-dimensional Hecke C^*-algebra $B_n \subseteq \mathcal{D}_H$ with identity e_H.

Because p_H is central, d_n commutes with p_H, and the spectra of $p_H d_n$ and $(e_H - p_H)d_n$ lie respectively near 1 and 0. For large n these clusters are separated by the circle Γ centered at 1 with radius 1/3. The spectrum of d_n in B_n equals its spectrum in $\mathcal{A}_H$: if an eigenvalue in the finite-dimensional algebra were absent from the corner spectrum, the corner inverse would annihilate its nonzero spectral projection. Consequently the Riesz projection

$$r_n = \frac{1}{2\pi i} \int_\Gamma (z e_H - d_n)^{-1}\, dz$$

belongs to $B_n \subseteq \mathcal{D}(G)$. Continuity of Riesz projections gives $r_n \to p_H$ in the $\mathcal{A}$-norm.

If $\ker \pi \in C$ and $\pi(e_H) \neq 0$, then $\pi(d_n) \to \pi(p_H) = 0$, so the Riesz projection of $\pi(d_n)$ around 1 is zero for large n. If $\pi(e_H) = 0$, every corner element is already annihilated. Hence $r_n \in \mathcal{D}(G) \cap \ker(C)$. Lemma 2 gives $r_n \in j(C)$, and closedness gives $p_H \in j(C)$. □

Theorem 5 (Clopen synthesis)**.** *Every clopen subset of* $\mathrm{Prim}_*(\mathcal{A})$ *is a set of synthesis.*

Proof. Let $f \in \ker(C)$ and fix compact-open H. Put $f_H = e_H * f * e_H$. Lemma 3 and separation by irreducible representations give

$$f_H = p_H * f_H.$$

Indeed both sides vanish on C; off C, p_H is the identity in every nonzero corner representation. Lemma 4 now gives $f_H \in j(C)$.

As H shrinks to the identity, Carter's compact-neighbourhood approximate identity [1, Proposition 2.16(ii)] yields $f_H \to f$ in $\mathcal{A}$. Therefore $f \in j(C)$. Since Carter already gives $j(C) \subseteq \ker(C)$, equality follows. □

6 Finite admissible synthesis

Call an irreducible π *admissible* if $\dim \mathcal{H}_\pi^H < \infty$ for every compact-open H.

Theorem 6 (Finite admissible synthesis)**.** *Every finite closed subset of* $\mathrm{Prim}_*(\mathcal{A})$ *consisting of admissible points is a set of synthesis.*

Proof. Let $C = \{\ker \pi_1, \ldots, \ker \pi_r\}$ and $I = \ker(C)$. For fixed H, evaluation maps $\mathcal{A}_H = e_H \mathcal{A} e_H$ into the finite-dimensional algebra

$$\bigoplus_{i=1}^{r} \mathcal{B}(\mathcal{H}_{\pi_i}^H).$$

The dense subspace $\mathcal{D}_H$ has the same image because a linear subspace of a finite-dimensional space is closed. Choose a bounded linear right inverse R_H from that image into $\mathcal{D}_H$.

For $f \in I$, approximate $f_H = e_H * f * e_H$ by $d_n \in \mathcal{D}_H$ and set

$$h_n = d_n - R_H\big((\pi_1(d_n), \dots, \pi_r(d_n))\big).$$

Then $h_n \in \mathcal{D}(G) \cap I$ and $h_n \to f_H$. Lemma 2 gives $f_H \in j(C)$, and the compact-open approximate identity gives $f \in j(C)$. □

Corollary 7 (p-adic unipotent orbit spaces). *Let N be a group of unipotent matrices over a p-adic field and let $\mathcal{A} = L^\Phi(N, \omega)$ satisfy the hypotheses above. Every finite or clopen subset of*

$$\mathfrak{n}^* / \operatorname{Ad}^*(N) \cong \operatorname{Prim}_*(\mathcal{A})$$

is a set of synthesis.

Proof. Carter's Theorem 8.10 states that N is CCR and identifies its unitary dual with the coadjoint-orbit space. For compact-open H, the projection $\pi(e_H)$ lies in $\pi(C^*(N)) = \mathcal{K}(\mathcal{H}_\pi)$, hence has finite rank. Thus every irreducible is admissible. CCR also makes primitive ideals maximal, so finite subsets are closed. Apply Theorems 5 and 6, together with Carter's Corollary 8.11. □

7 Sharpness: compact closed sets can fail synthesis

Theorem 8 (Malliavin obstruction). *Let k be a non-archimedean local field and identify $U_2(k) \cong (k, +)$. For the unweighted Hermitian algebra $\mathcal{A} = L^1(k)$, there is a compact closed set*

$$C \subseteq \operatorname{Prim}_*(\mathcal{A}) \cong \widehat{k} \cong k$$

which is not a set of synthesis. Consequently, neither closedness nor compactness alone can replace finiteness or clopenness in Corollary 7.

Proof. The case $\Phi(t) = |t|$ and $\omega = 1$ gives $\mathcal{A} = L^1(k)$ and lies in the source framework. Malliavin proved that spectral synthesis fails for $L^1(G)$ whenever the locally compact abelian group G is noncompact [2]. In its compact-set formulation, every nondiscrete locally compact abelian dual contains a compact subset on which synthesis fails [3, p. 358]. The additive group of k is noncompact and Pontryagin self-dual, so this gives a compact closed $C \subseteq \widehat{k}$ and a closed ideal I satisfying

$$\operatorname{hull}(I) = C, \qquad I \subsetneq \ker(C).$$

Carter's Theorem 7.1 makes $j(C)$ the minimal closed ideal for this hull, so $j(C) \subseteq I \subsetneq \ker(C)$. Hence $j(C) \neq \ker(C)$. □

8 Verification, limitations, and novelty

No numerical computation is used. The proof audit checks the Hermitian corner, the complementary-kernel central idempotent, equality of spectra in the finite Hecke algebra and the corner, Riesz-projection convergence, and the fact that no uniform finite-corner lift is needed as H shrinks.

The theorem does not classify infinite closed subsets with nonempty boundary. Theorem 8 shows that this is a genuine obstruction rather than merely a limitation of the proof: even compact closed subsets can fail synthesis in the abelian p-adic case. In a non-Hausdorff primitive spectrum, "compact-open" alone need not imply closed; Carter's $j(C)$ is defined for closed C, while the clopen theorem has exactly the required closedness. Local indexes and bounded arXiv searches on 21 July 2026 combined the exact source id with "spectral synthesis", "p-adic unipotent", "clopen", "compact-open", "finite sets", "central idempotent", and "weighted Orlicz". No exact theorem was found; novelty confidence is moderate pending expert review.

Human-review recommendation. Verify Lemma 3's primitive corner correspondence and Lemma 4's equality of the finite-Hecke and corner spectra. Those are the decisive new structural steps.

References

[1] M. Carter, *Weighted Orlicz $*$-algebras on locally elliptic groups*, Studia Math. **287** (2026), no. 2, 145–180; arXiv:2503.20735v2.

[2] P. Malliavin, *Impossibilité de la synthèse spectrale sur les groupes abéliens non compacts*, Publ. Math. IHÉS **2** (1959), 61–68.

[3] K. de Leeuw and H. Mirkil, *Translation-invariant function algebras on abelian groups*, Bull. Soc. Math. France **88** (1960), 345–370.